

DC-DC Buck Converter Report B2-G5

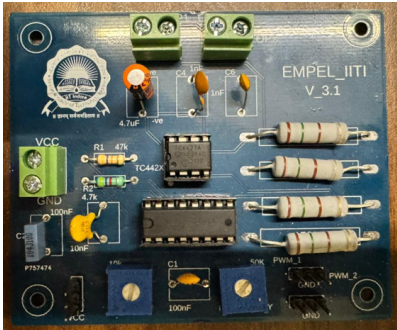
Roll No.	Name
240002049	Parekh Om Nirav
240002064	Praharsha Sandineni
240002071	Siddhartha Porika
240002082	Yash Brijesh Modi

Objective: Design and analyze a DC-DC Buck Converter using TL494 and observe CCM and DCM modes.

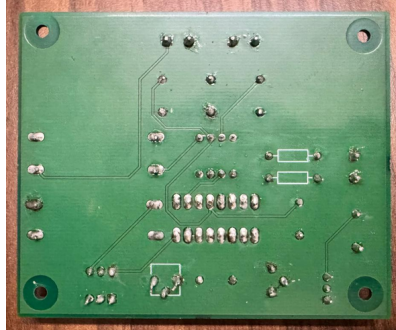
Specifications: $V_{in} = 20\text{ V}$, $V_{out} = 7.5\text{ V}$, $f = 20\text{ kHz}$, $I_{out} = 1\text{ A}$

Drive Link: https://drive.google.com/file/d/1GbvmQU773Ghvoxdu7S2UesZ6Kj_01g1M/view

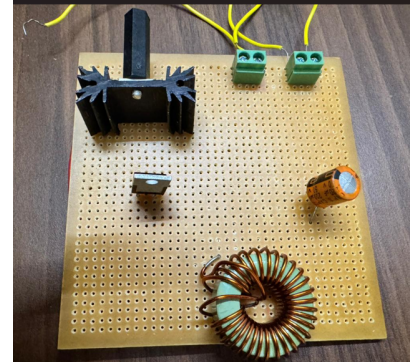
Hardware



(a) Gate Driver (Front)



(b) Gate Driver (Back)



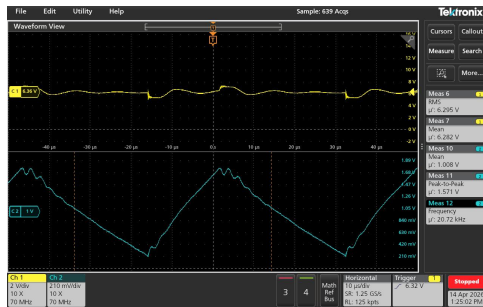
(c) Converter

Gate Pulse



Gate Pulse 20 kHz, Duty Cycle 37.5%

CCM

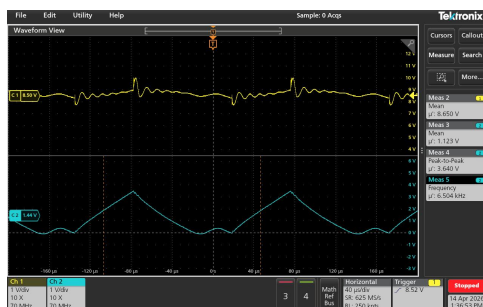


(a) Output Voltage = 6.3V, Output Current = 1 A, Frequency = 20.7 kHz



(b) Switch Voltage ($V_{avg} = 17.87\text{ V}$) and Inductor Current ($I_{avg} = 970\text{ mA}$)

DCM



(a) Output Voltage = 8.65 V, Frequency = 6.5 kHz



(b) Switch Voltage ($Avg = 15.20\text{ V}$) and Inductor Current ($Avg = 1.8\text{ A}$)

Conclusion: The Output Voltage of Buck Converter is given by $V_o = DV_{in}$.